

DATA TYPES

Data type determines the type of value we are going to store in our computer. To store anything in our computer, we should have to allocate the memory. This memory allocation is depended on the data type we are using.

Data type determines the properties such as

1. No of bytes
2. Range
3. Type of value

In C language we are having 3 **basic** data types

- 1. Int – To store non-decimal numbers**
- 2. Float – To store decimal numbers**
- 3. Char – To stores alphabets, numbers and special char**

Total data types are divided into 3 types.

- 1. Primitive data types**
- 2. Derived data types**
- 3. User defined data types**

PRIMITIVE DATA TYPES:

These are the regular data types we are using in our c programs.

Data type	Bytes	Conversion Character / format specifier	Storage Range
int / signed int / short int	2	%d	-32768 to +32767
unsigned int	2	%u	0 to 65535
long int	4	%ld	-2147483648 to 2147483647
unsigned long int	4	%lu	0 to 4294967295
float	4	%f	$3.4 * 10^{-38}$ to $3.4 * 10^{+38}$
double	8	%lf	$1.7 * 10^{-308}$ to $1.7 * 10^{+308}$
long double	10	%Lf	$3.4 * 10^{-4932}$ to $1.1 * 10^{+4932}$
char	1	%c	1 character Signed char [-128 to +127] Unsigned char [0 to 255]
char[10] (STRING)	10	%s	9 char + 1 null char
void [empty data type]			nothing

DERIVED DATA TYPES:

They are derived from primitive data types.

1. Array [non-primitive]
2. Pointer

3. Function

USER DEFINED DATA TYPES:

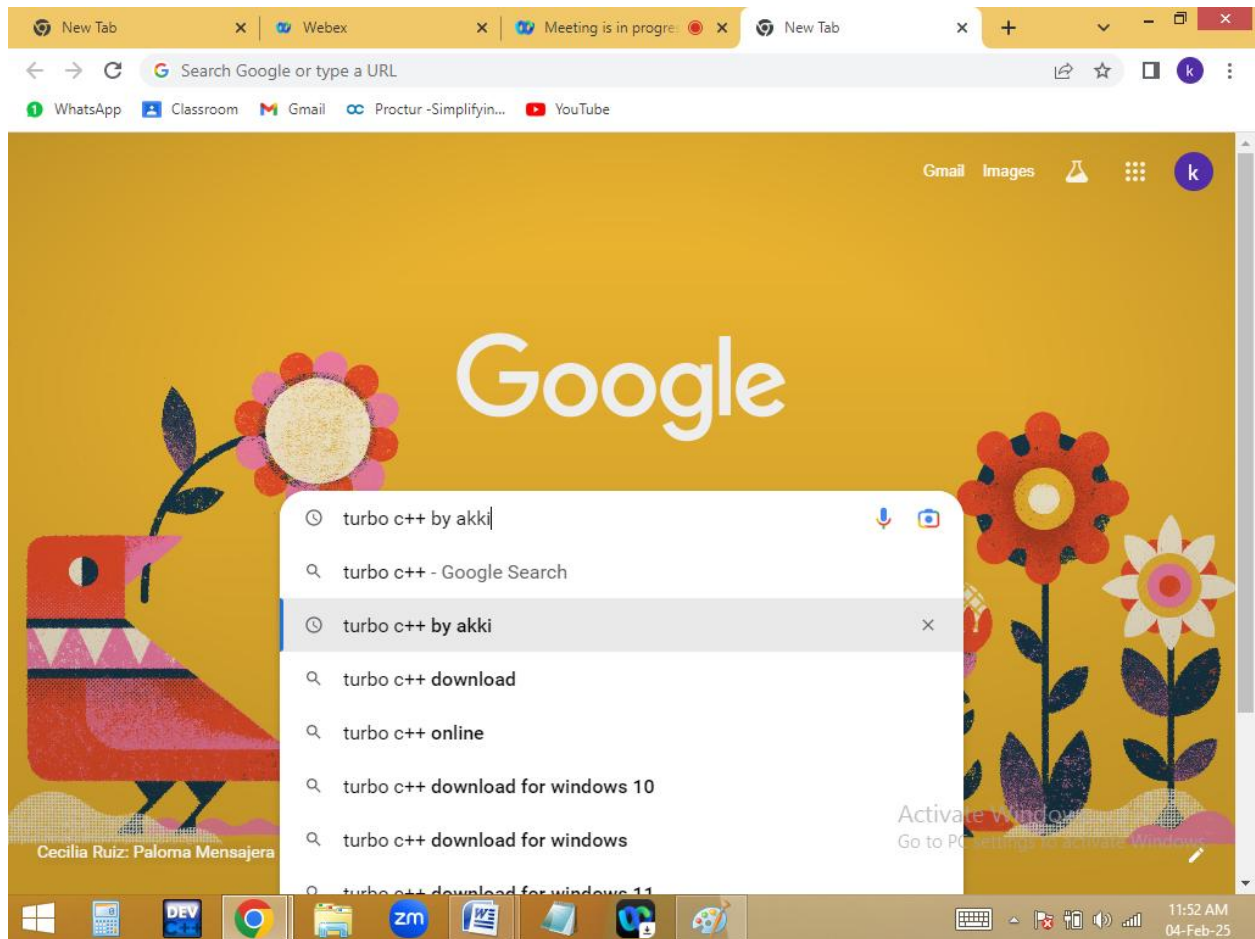
These are the data types created by the user.

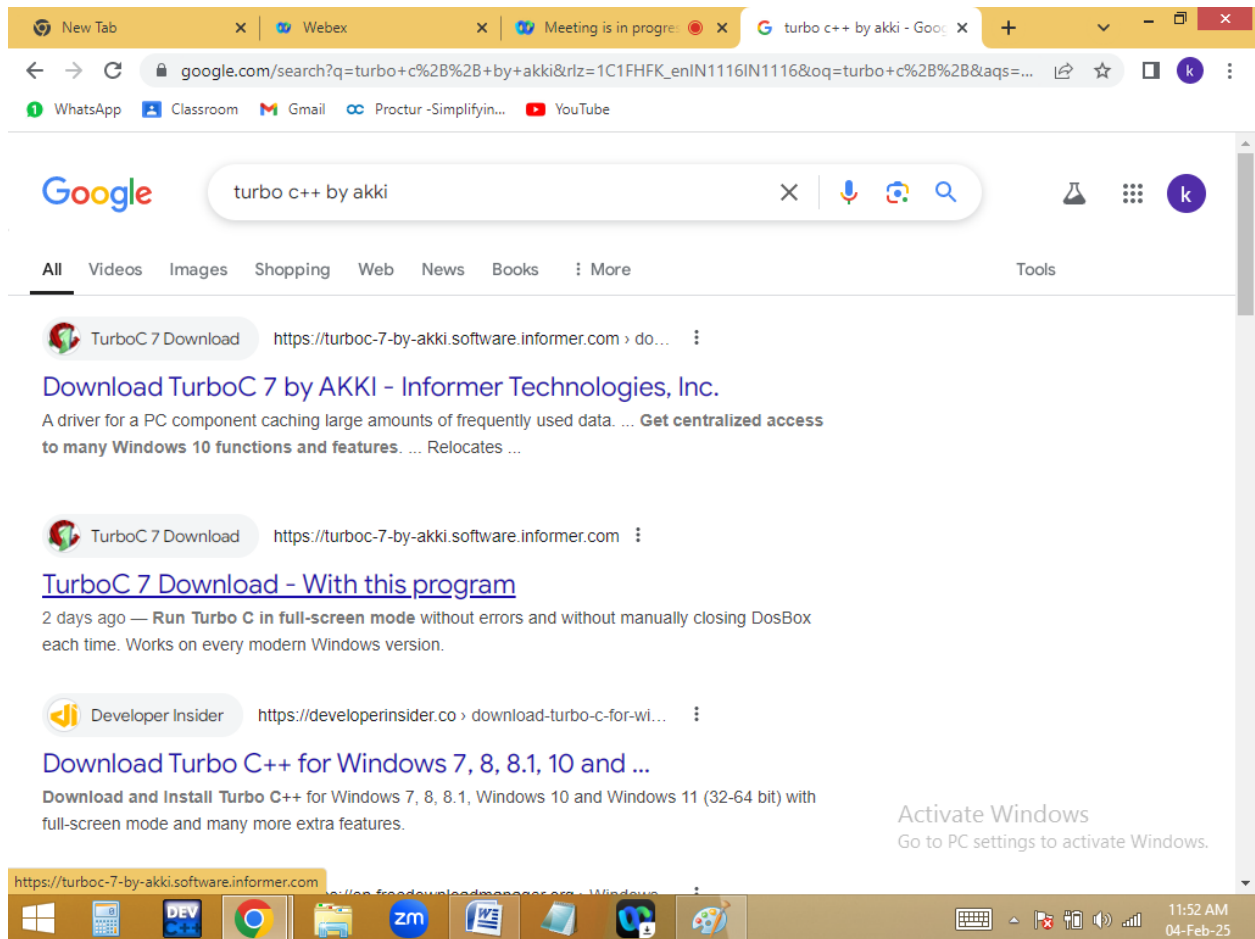
1. structure
2. union
3. enum

Installation of C++ software:

Open a browser [chrome]

Type turbo c++ by akki





Click on download turboc 7 by AKKI

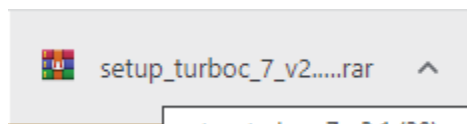
Or

Click on below link using Ctrl+click

<https://turboc-7-by-akki.software.informer.com/download/>

Now it downloads the turbo c++ automatically.

Click on

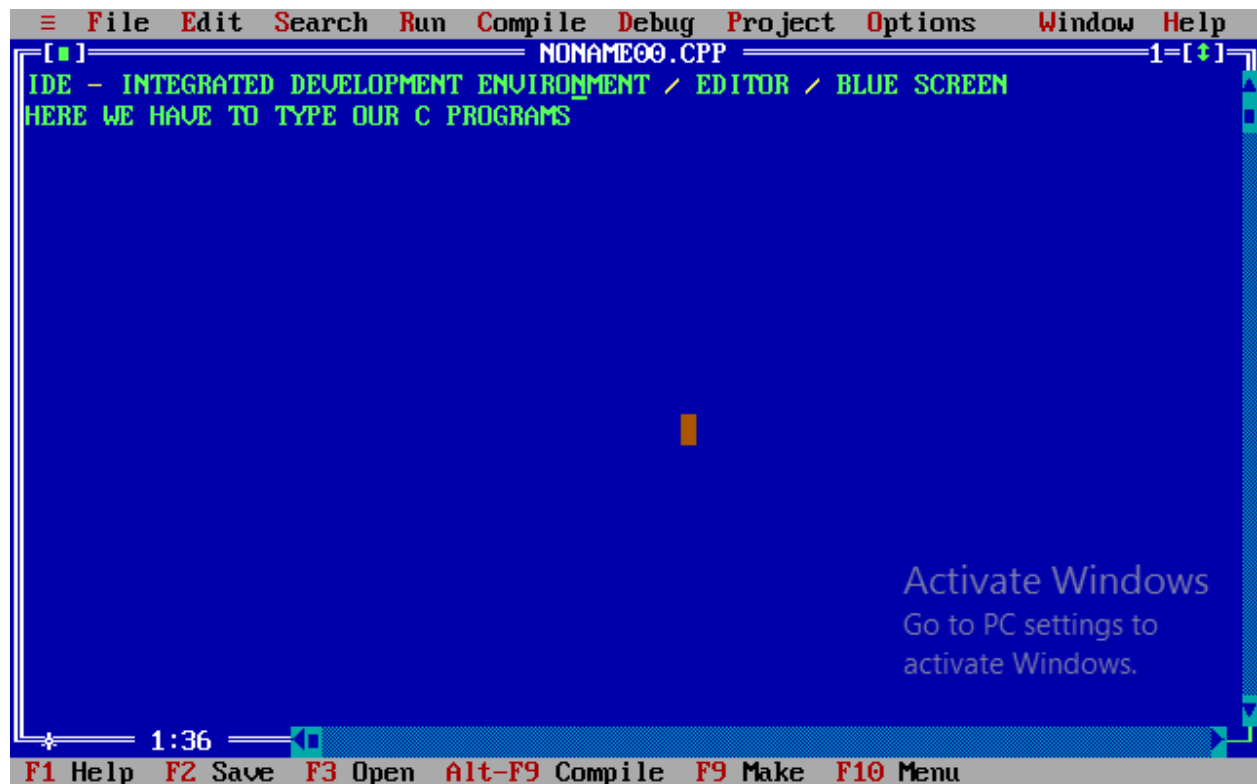


Note: we have to install winrar / unzip / 7 zip etc before installation

Double click on Setup_TurboC_7_v2.1.exe

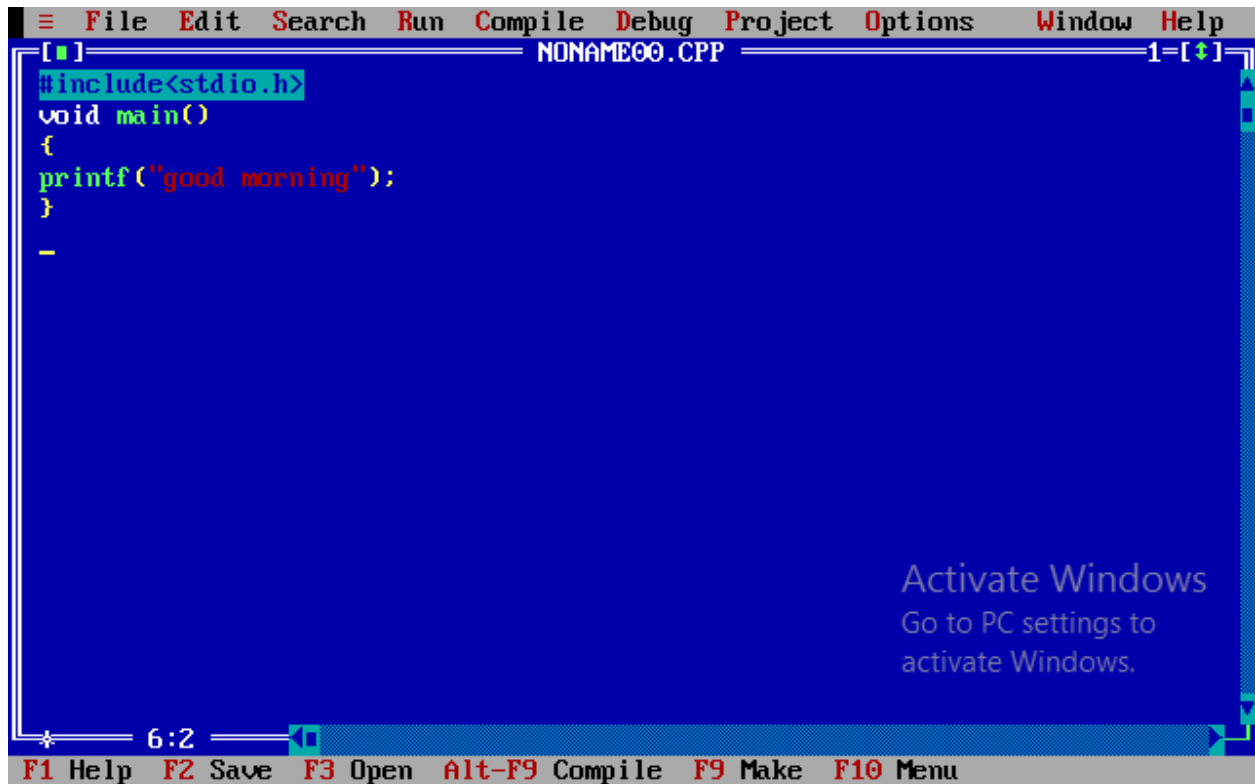
Click on yes → next → next → next → next → install → finish

Finally we are getting a blue window as follows.



To get full / half screen press Alt+Enter

Sample program?



The screenshot shows the Turbo C++ IDE interface. The menu bar at the top includes File, Edit, Search, Run, Compile, Debug, Project, Options, Window, and Help. The title bar indicates the file is NONAME00.CPP. The code editor contains the following C program:

```
#include<stdio.h>
void main()
{
printf("good morning");
}
-
```

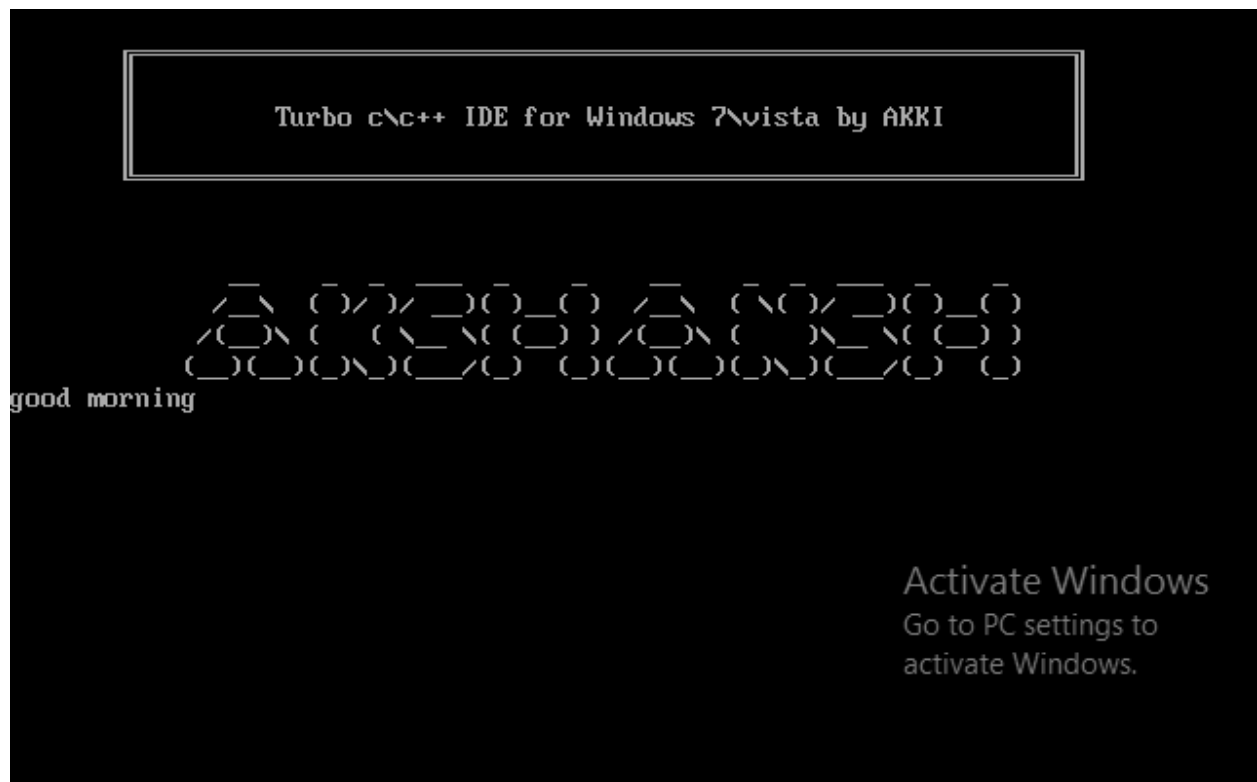
The status bar at the bottom shows the cursor position 6:2 and a list of function key shortcuts: F1 Help, F2 Save, F3 Open, Alt-F9 Compile, F9 Make, and F10 Menu. A watermark in the bottom right corner reads "Activate Windows Go to PC settings to activate Windows."

Save the file - press F2 or Alt+F and select save [in laptop Fn + F2]

Compile the program – Alt+F9 or Alt+C and select compile

Run – Ctrl+F9 or Alt+R and select Run

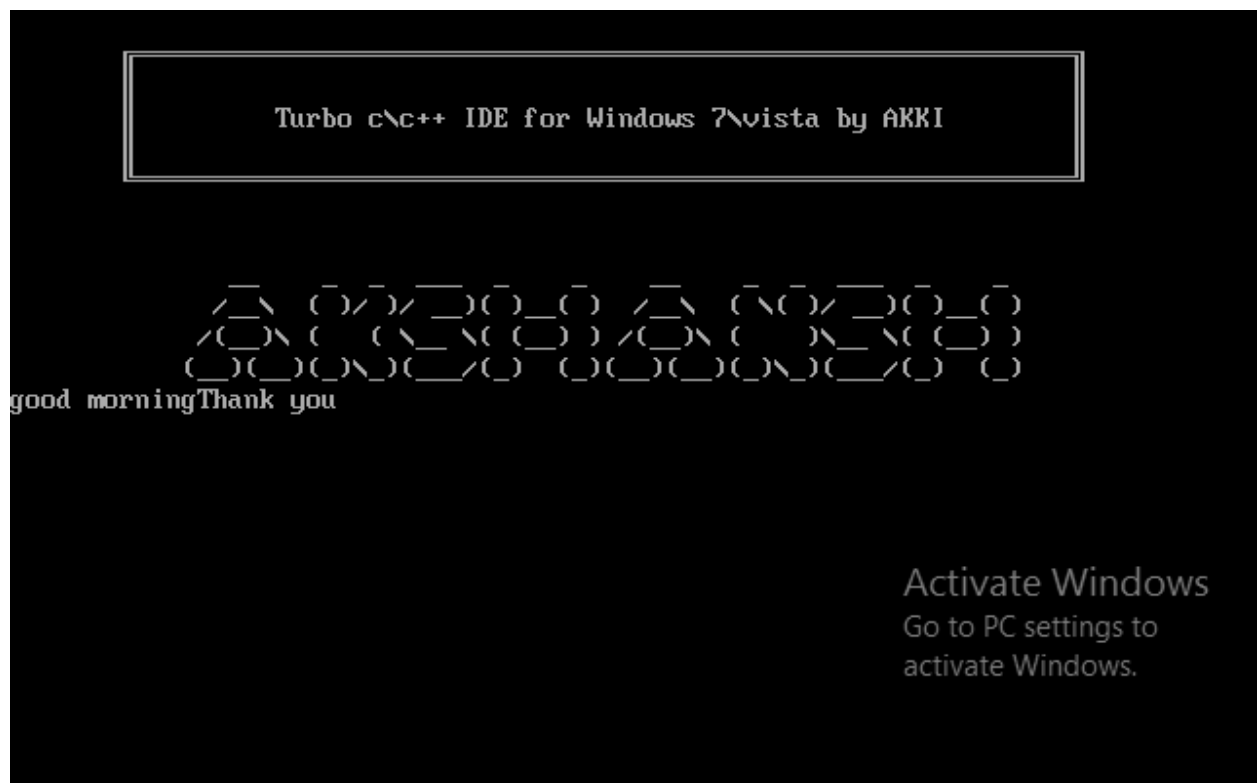
Output window – Alt+F5 or Alw+W and select User screen

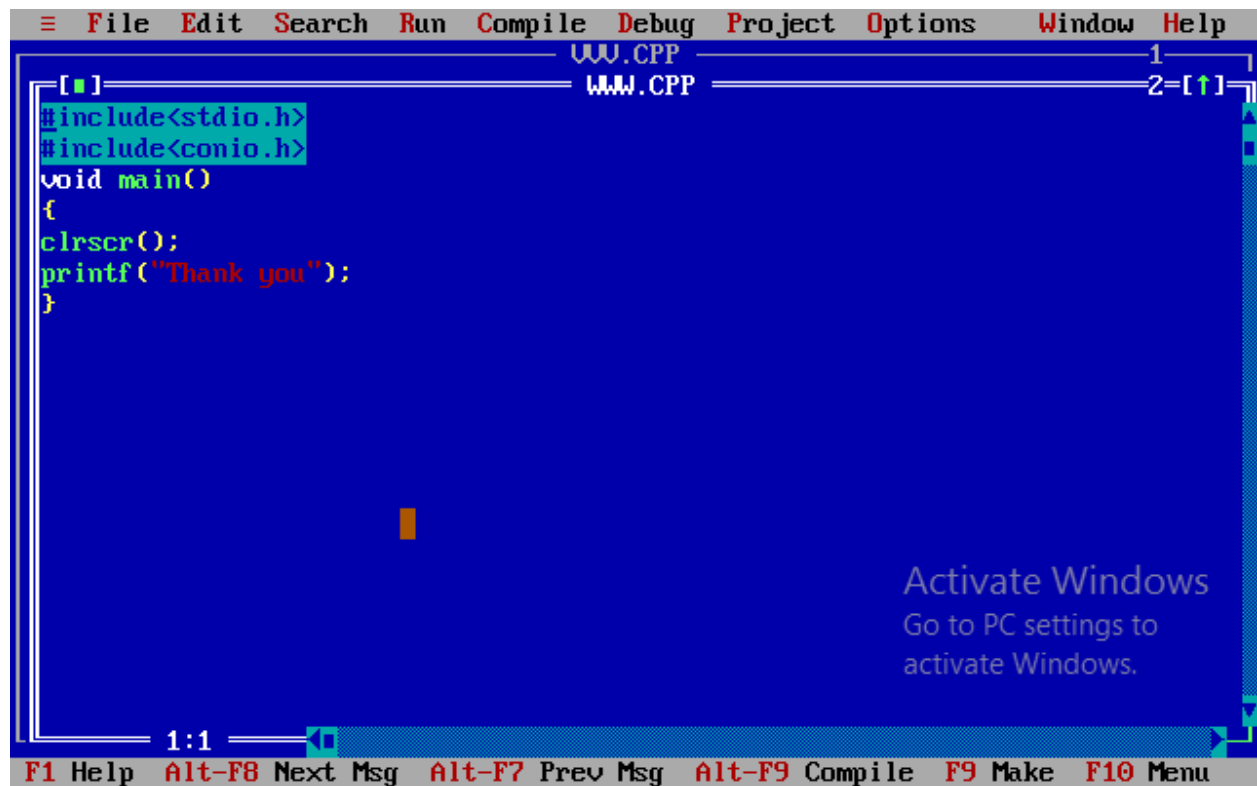


This black window is called

Command prompt / console / run window / user screen /
output window

To create a new program click on file menu and select
new

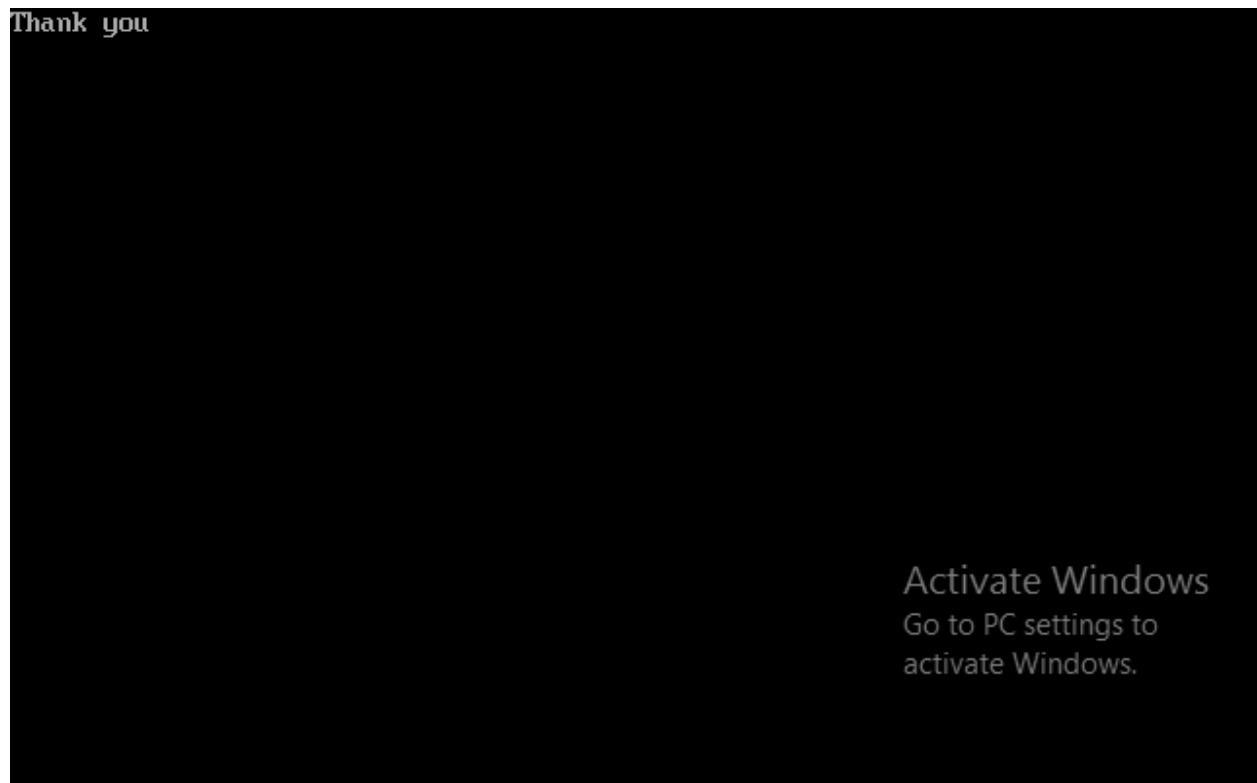




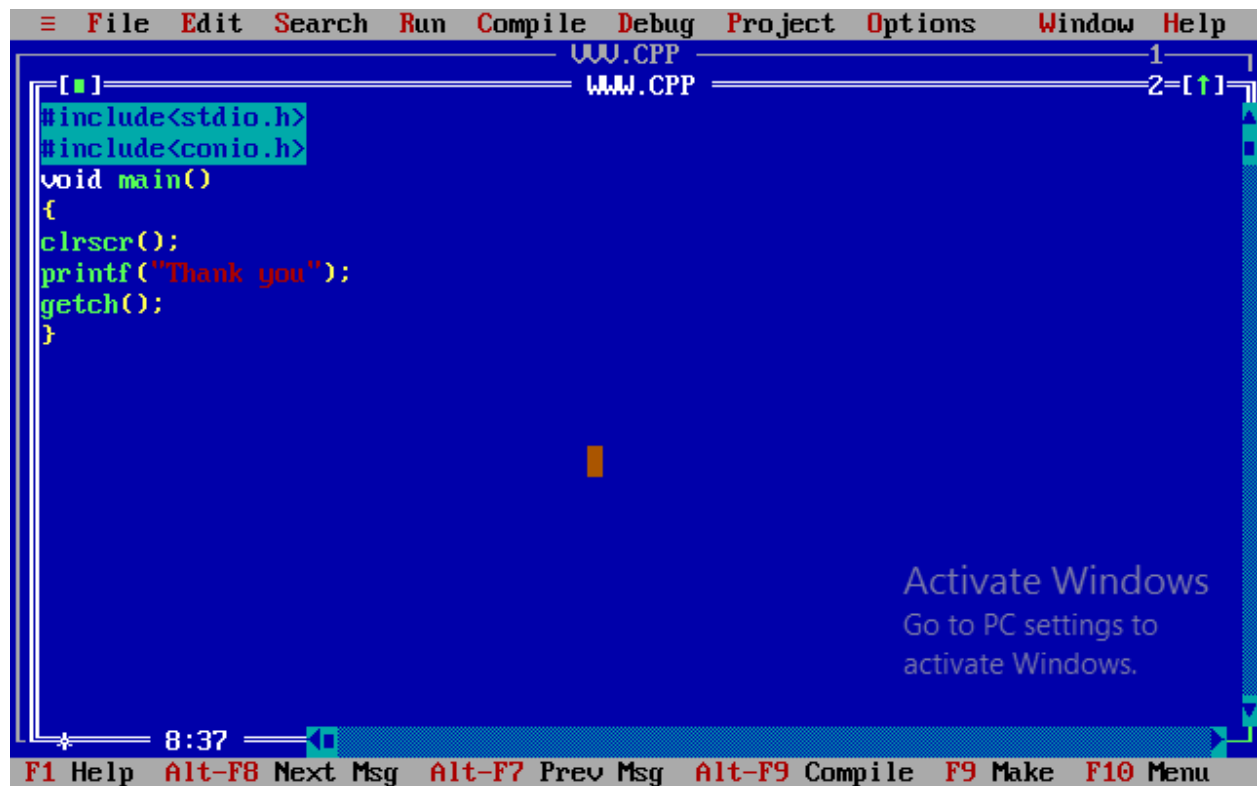
The screenshot shows the Turbo C++ IDE interface. The menu bar at the top includes File, Edit, Search, Run, Compile, Debug, Project, Options, Window, and Help. The window title bar indicates the file is 'WWW.CPP'. The code editor contains the following C++ code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Thank you");
}
```

The status bar at the bottom displays function key shortcuts: F1 Help, Alt-F8 Next Msg, Alt-F7 Prev Msg, Alt-F9 Compile, F9 Make, and F10 Menu. A watermark in the bottom right corner reads 'Activate Windows Go to PC settings to activate Windows.'



The screenshot shows a black terminal window with the text 'Thank you' printed in white at the top left. A watermark in the bottom right corner reads 'Activate Windows Go to PC settings to activate Windows.'



The screenshot shows the Turbo C++ IDE interface. The menu bar at the top includes File, Edit, Search, Run, Compile, Debug, Project, Options, Window, and Help. The window title bar indicates the file is 'WWW.CPP'. The code editor contains the following C++ code:

```
[ ]  
#include<stdio.h>  
#include<conio.h>  
void main()  
{  
clrscr();  
printf("Thank you");  
getch();  
}
```

The status bar at the bottom displays the time '8:37' and various function key shortcuts: F1 Help, Alt-F8 Next Msg, Alt-F7 Prev Msg, Alt-F9 Compile, F9 Make, and F10 Menu. An 'Activate Windows' watermark is visible in the bottom right corner of the IDE window.

Thank you_

Activate Windows
Go to PC settings to
activate Windows.

To open old file F3 or Alt+F and select open. Select the file enter key

To close the current file click on green box or alt+F3.

To close c++ Alt+X or Alt+F and select Quit.

To close all the files at a time Alt+W and select Close all.

F5 for full blue window.

F6 – switch between files

OPERATORS

Operator is a special symbol is used to do a particular task. In c we are having 44 operators and 14 separators.

Operator works on operands. Based on no of operands participating in operation, the operators divided into 3 types.

1. Unary operators: Require one operand

Eg: `a++`, `a--`, `+a`, `-a`, `!a`, `~a`, `sizeof(a)`,.....

2. Binary operator: Require two operands.

Eg: `a+b`, `a*b`, `a>b`, `a==b`, `a!=b`, ...

3. Ternary / conditional op [? :]: It require 3 operands.

Conditional part ? true part : false part;